Internet Protocol TV (Credit: www.techybugz.com)

control. The separated-out video and audio information consist of three signals—composite video, audio L and audio R—which are interfaced to the display unit.

Digital cable TV (DVB-C/C2)

Digitisation of cable TV has also been implemented in a similar way. Digital bit-stream, in case of DVB-C/C2, is similar to the one used in case of DVB-S/S2 standard. Quadrature amplitude modulation (QAM) is used for modulating the signal onto an RF carrier in 47MHz - 862MHz frequency range. DVB-C/C2 differs only as much that, input RF received from the cable is in VHF/lower UHF range—rest of the processing is similar to that described in case of DVB-S/S2.

Digital terrestrial TV (DTT)

Digital Video Broadcasting - Terrestrial (DVB-T/T2), utilises VHF/UHF carriers in 174MHz - 862MHz range, and can transmit eight sub-channels in the space used by 8MHz analogue TV channels. In case of DTT, signals are transmitted as data compressed digital bit-stream modulated on a carrier in UHF band, similar to the ones allotted earlier to analogue UHF TV. Since signal is prone to multi-path and fading, orthogonal frequency division multiplexing (ODFM) is used to transform digital bit-stream to a frequency multiplex.

In case of DTT, a simple UHF Yagi antenna is used for signal reception. An STB, designed for DVB-T/T2 operation, is used to recover and decode digital bit-stream, and reconstruct video as well as audio. Since DTT requires a simple installation, it may be handled by viewers themselves.

Doordarshan is broadcasting at present five sub-channels in one digital channel. Currently, DD National, DD News, DD Bharati, DD Sports and DD Regional/DD Kisan are supported on one channel. Channel frequencies used in India are given in the table. Frequency range extends from 470MHz - 562MHz. The ~ 100MHz bandwidth required in the antenna can be easily achieved in a suitable Yagi antenna. TV sets capable of handling DVB-T/T2 standard are a few at present.

Internet Protocol TV (IPTV)

Internet Protocol TV is rather different. Typically, an Internet connection is made available to the telephone subscriber with the help of a signal splitter to separate low-frequency audio signal used in telephony, and the high-frequency phase modulated sub-carrier. The copper connection to the telephone subscriber's public switched telephone network (PSTN) is capable of transmitting high-frequency signals apart from low-frequency signals needed for telephony. This property is used to realise advanced digital subscriber line (ADSL) connection.

Data bit-stream, available in an Internet connector, is decoded, and video and audio information is separated. This is performed in a specific STB, under program control. The unique feature here is that video signal is accommodated within the pass band available in the telephone line by data compression.

Signal recovery at the user end requires processor power to accomplish signal separation and decompression. Unique feature in case of IPTV is that, the user sees only one programme at a time, and connection to the exchange is bi-directional. Therefore the user can send a command via remote control to select any programme of his or her choice. The service provider, in turn, directs data